

EESSI

EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

7 May 2026

https://github.com/EESSI/meetings/wiki

This meeting is being recorded

- Be aware: If you speak up, you will be in the recording
- Feel free to raise questions via Zoom chat instead
- We'll do informal Q&A at the end (after stopping recording)
- Recording will be made available via EESSI YouTube channel:

  https://www.youtube.com/@eessi_community

Agenda



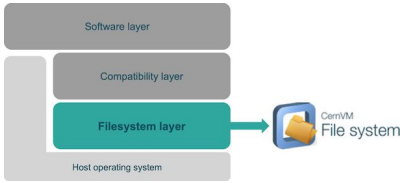
- Quick introduction by new people
- Progress update per EESSI layer
- Update on EESSI production repository `software.eessi.io` (+ RISC-V)
- Update on EESSI test suite, build-and-deploy bot, documentation, etc.
- AWS sponsorship update
- EESSI governance
- Upcoming/recent events
- Q&A

Quick introduction by new people

New people on the call: feel free to introduce yourself!

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

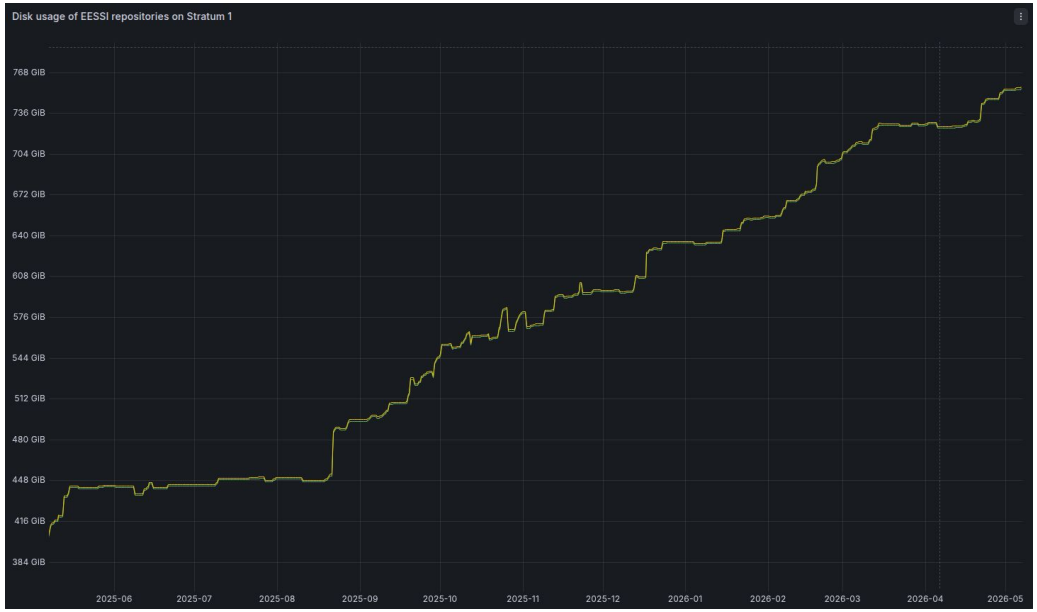
Progress update: filesystem layer



- Added an EESSI client container for macOS 26 (PR #266)
  - See also https://eessi.io/docs/getting_access/eessi_Apple_container_tool
- No issues with the CernVM-FS infrastructure, all servers current have:
  - Latest kernel version, incl. patch for Copy Fail
  - CernVM-FS version 2.13.1
- TODO: new release of `cvmfs-config-eessi` package to configure CernVM-FS
  - Including info on recently added Stratum-1 mirror server hosted by RAL (UK)
  - Including new options, e.g. to use a CDN (http://s1eessieu1-cvmfs.openhtc.io, etc.)

[Bob, Kenneth]

Progress update: filesystem layer monitoring

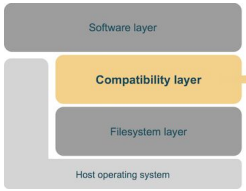


Size is (still) growing with ~1GB/day

Currently ~750GB of disk space is required
for a full Stratum-1 mirror of all EESSI repositories

[Bob, Kenneth]

Progress update: compatibility layer

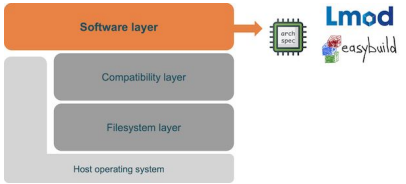


- User resolving may be broken if `libnss_db` is used on SLES-based systems
  - https://github.com/EESSI/compatibility-layer/issues/234
  - Database files are in `/var/lib/misc` instead of `/var/db`
  - Should we use variant symlinks for such paths?

- Open issue in EESSI support portal to collect ideas for the 2026.x compatibility layer: https://gitlab.com/eessi/support/-/issues/220
  - Effort on this hopefully starting soon…

[Bob, Kenneth]

Progress update: software layer (1/2)



Highlights of recently merged software PRs (2023.06):

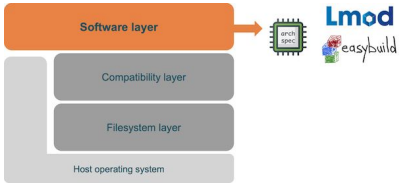
- 655 distinct software packages available now,
  4 merged PRs since last update meeting (5 Mar'26)

Highlights of recently merged software PRs (2025.06):

- 437 distinct software packages now available,
  30 merged PRs since last update meeting (5 Mar'26)
- More top level applications available:
  LAMMPS vJul2025 (PR #1335), AmberTools 25.2 (PR #1457),
  Score-P v9.4 + Scalasca v2.6.2 (PR #1467), imageio v2.37.0 (PR #1468),
  scikit-image v0.25.0 (PR #1471), GROMACS 2025.4 (PR #1484)
- CUDA 12.9.1 and cuDNN 9.15.0.57 (PR #1410)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo, Helena …]

Progress update: software layer (2/2)

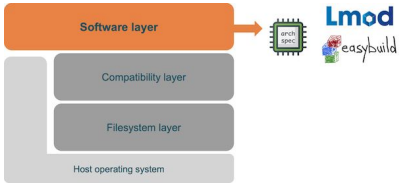


Rebuilds of existing software installations:

- EESSI 2023.06:
  - RStudio Server 2024.09 (PR #1469)
  - FFmpeg 6.0 (PR #1490)
- EESSI 2025.06
  - Rebuild FFmpeg for zen5 CPU target (PR #1447)
  - RStudio Server 2025.12 (PR #1470)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

Progress update: software-layer-scripts



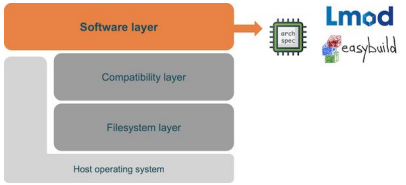
Highlights of recently merged software-layer-scripts PRs:

- Several PRs for AMD GPU support (PR #206, PR #215, PR #217, PR #218)
- `eessi_archdetect.sh`: Allow for a default vendor fallback for ARM CPUs (PR #173)
- Script for reproducing an existing stack for a new CPU target (PR #163)
- Drop support for compute capability <= 7.0 for newer cuDNN versions (PR #170)

+ various other minor PRs: #136, #172, #175, #176, #179, #180, #182, #185, #191, #192, #193, #194, #195, #196, #198, #199, #200, #202, #203, #204, #207, #209, #211, #212

[Kenneth, Alan, Bob, Richard, Thomas …]

Progress update: RISC-V



RISC-V software stack being built via https://github.com/EESSI/dev.eessi.io-riscv

- See page in documentation: https://www.eessi.io/docs/repositories/dev.eessi.io-riscv/
- Installations targeting `/cvmfs/dev.eessi.io/riscv` repo
  - Same installation workflow through bot
  - Built on top of EESSI 2025.06 compatibility layer for RISC-V (available in `software.eessi.io`)
- >300 (+125 since last meeting) installations have already been ingested 🎉
  - `foss/2025a`, `foss/2025b`, and **`lfoss/2025b`** are available
- Planning to also look into:
  - More end-user applications
  - Providing an overview of available software in the documentation
    - https://github.com/EESSI/docs/pull/713
  - Going beyond `riscv64/generic` as target, also optimize for Banana Pi (RISC-V CPU w/ vector instructions)

[Bob, Julian, Kenneth, …]

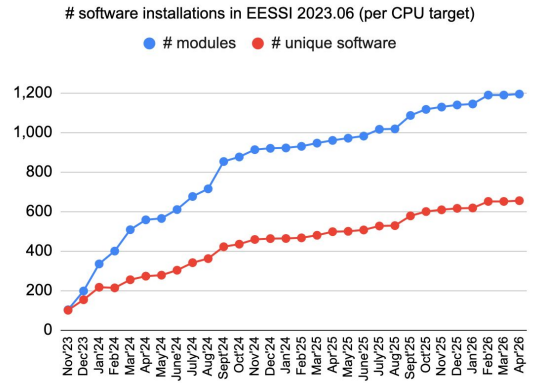
EESSI production repository

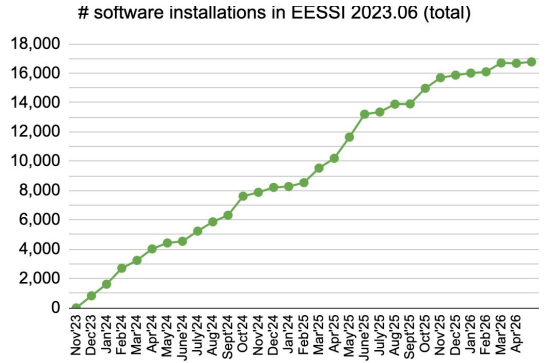
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2023.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 14 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 1,194 software installations per CPU target**
  (+8 in Mar+Apr'26)
  - **655 different software projects** (+4 in Mar+Apr'26)
    16,779 software installations (across 14 CPU targets, +700 in Mar+Apr'26)
- Current efforts:
  - Adding more GPU (CUDA) software installations
  - Processing incoming contributions
- See overview of available software: https://eessi.io/docs/available_software
- **Attention is shifting to EESSI version 2025.06 (see next slide)**





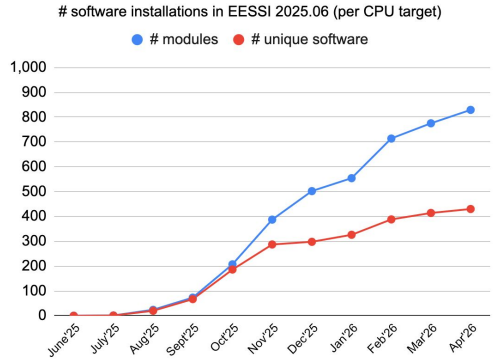
EESSI production repository

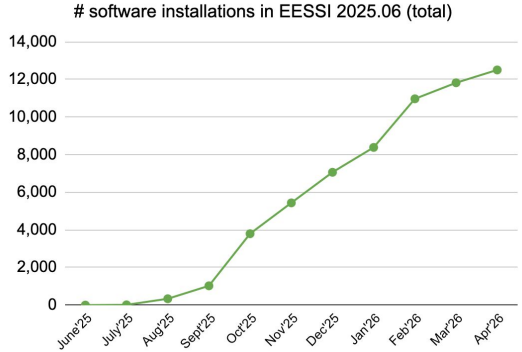
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2025.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- **15** fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- **Extra CPU target was added: AMD Zen 5 (Turin)**
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 829 software installations per CPU target** (+115 in Mar+Apr'26)
  - **430 different software projects** (+42 in Mar+Apr'26)
    12,505 software installations (across 15 CPU targets, +1,538 in Mar+Apr'26)
- See overview of available software: https://eessi.io/docs/available_software
- Current focus:
  - Add CUDA installations
  - Adding more software, processing incoming contributions
- Coming soon (hopefully)
  - Add support for Graviton 4+5





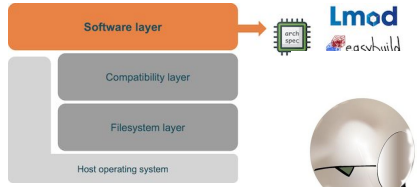
Support for AMD Zen 5 (Turin)



- Support for `x86_64/amd/zen5` has been added to EESSI 2025.06
  - Using eessi_software_reproduce_stack.py script to reproduce the zen2 stack
  - No big issues encountered: https://github.com/EESSI/software-layer/issues/1430
  - Accelerator stacks are also available
- Various tests and benchmarks have been run
  - Quite a lot of differences in reported CPU flags between different Zen 5 nodes
  - Does not seem to have impact on performance
- Zen5 auto-detection will soon be merged into archdetect (see PR #162)
- Try it out already:

```
export EESSI_SOFTWARE_SUBDIR_OVERRIDE=x86_64/amd/zen5
source /cvmfs/software.eessi.io/versions/2025.06/init/lmod/bash
```

Bot for building + deploying software





https://github.com/EESSI/eessi-bot-software-layer + https://github.com/EESSI/filesystem-layer

Current bot version 0.11.0 has been released (28 Jan'26) release notes

Developments since then

- [MERGED] added unit tests to increase test coverage (PR #365)
- [MERGED] more robust parsing of scontrol output (PR #369)
- [REVIEWED] modeling GitLab events (PR #370)
- [OPEN] don't check export variables in check_filter (PR #364)

[Bob, Caspar, Kenneth, Lara, Pedro, Sam, Sondre, Thomas]

Improved overview of available software

eessi.io/docs/available_software

(in place since end of Feb'26)

Software available in EESSI

Overview of software available in EESSI's production repository `software.eessi.io`.

752 unique software projects (+ 3007 unique extensions)

name:gromacs

GROMACS

(more details)

https://www.gromacs.org

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles. This is a CPU only build, containing both MPI and threadMPI binaries for both single and double precision. It also contains the gmxapi extension for the single precision MPI build.

Available in EESSI versions: 2023.06 2025.06

Supported CPU families: AMD Intel Arm

Supported GPU families: NVIDIA

GROMACS

GROMACS is a versatile package to perform molecular dynamics, i.e. simulate the Newtonian equations of motion for systems with hundreds to millions of particles.

This is a GPU enabled build, containing both MPI and threadMPI binaries.

It also contains the gmxapi extension for the single precision MPI build.

homepage: https://www.gromacs.org

Available installations

GROMACS version	Supported CPU targets	Supported GPU targets	EESSI version	Module
2025.4	generic: aarch64, x86_64 Arm: a64fx, neoverse_n1, neoverse_v1, nvidia/grace AMD: zen2, zen3, zen4 Intel: haswell, skylake_avx512, sapphirerapids, icelake, cascadelake	*(none)*	2025.06	GROMACS/2025.4-foss-2025b
2025.2	generic: aarch64, x86_64 Arm: a64fx, neoverse_n1, neoverse_v1, nvidia/grace AMD: zen2, zen3, zen4 Intel: haswell, skylake_avx512, sapphirerapids, icelake, cascadelake	*(none)*	2025.06	GROMACS/2025.2-foss-2025a
2024.4	generic: aarch64, x86_64 Arm: neoverse_n1, neoverse_v1, nvidia/grace AMD: zen2, zen3, zen4 Intel: haswell,	NVIDIA: cc70, cc80, cc90	2023.06	GROMACS/2024.4-foss-2023b-CUDA-12.4.0

EESSI documentation

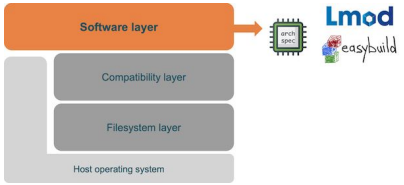
eessi.io/docs - GitHub repo github.com/EESSI/docs

Highlights:

- EESSI Events & Training
  - Added EESSI Webinar Series (+ added links to recordings)
  - Added EESSI Happy Hour (+ added links to recordings)
- Blog posts (https://www.eessi.io/docs/blog/)
  - Update on EESSI in EuroHPC Federation Platform
  - Choosing EESSI as a base for MUSICA
- EESSI via Apple container tool
- Use EasyBuild's `LooseVersion` to compare version numbers on software overview page
- Plus other minor changes to the documentation (software overview, corrections,...)

[Lara, Pedro, Helena, Bob, Xin, Kenneth, Caspar…]

EESSI test suite



eessi.io/docs/test-suite - github.com/EESSI/test-suite

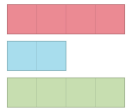
ReFrame

Highlights

- Add support for testing multiple EESSI versions in a single run, using the concept of ReFrame programming environments PR #326 (WIP)
- Number of fixes/improvements to existing suite:
  - Pull `hpctestlib` files from ReFrame into EESSI testsuite, as first step to stop using them since `hpctestlib` is not being developed further (PR #328)
  - Migrated last test (MetalWalls) to use the EESSI_Mixin class (PR #332)
  - Number of other fixes: PR #324, PR #327, PR #334, PR #336, PR #338, PR #313 (WIP)

[Caspar, Sam, Satish, Kenneth, Lara]

EasyBuild v5.3.0





Released on 10 April 2026, see https://pypi.org/project/easybuild

Includes:

- Support for ROCm-based toolchains (`rfoss` & co), but no easyconfigs included yet
- *Experimental* support for installing with bwrap (see `--bwrap` and `--bwrap-installpath`), to do staged software installations before deploying them (see also EUM'26 talk by Bart & Sam)
- Enhance `PythonPackage` and `PythonBundle` to verify Python package names and versions with `pip list` (only enabled by default when uploading test reports in easyconfig PRs)
- Add easyconfigs for update of (common) toolchains:
  `foss/2026.1`, `intel/2026.1`, `lfoss/2026.1`

[Kenneth]

Status update of EESSI on EuroHPC systems



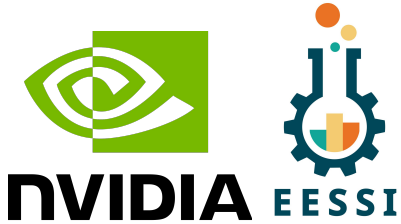


- EESSI is currently available on 8 out 9 current EuroHPC supercomputers:
  LUMI*, Vega, Karolina, Deucalion, MareNostrum 5, Discoverer, MeluXina, Leonardo*
- Groundwork to make EESSI available on JUPITER is done
- EESSI is expected to be available on all 9 (+ upcoming) EuroHPC supercomputers soon, thanks to EuroHPC Federation Platform (EFP)
  - Upcoming EuroHPC AI Factories will also be required to integrate with EFP
  - See also https://docs.my-eurohpc.eu/software-catalog/system-specific/about
- 1st release of EFP happened in April
  - Includes EESSI as base of Federated Software Catalog component (in beta)
  - More info on EFP: https://my-eurohpc.eu

(*) only available on-demand

[Kenneth, Alan, Lara]

Improved NVIDIA GPU support

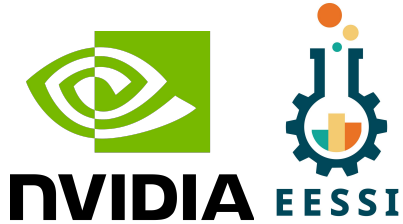


Software-layer:

- Added additional CUDA packages
  - CUDA 12.9.1 + OSU + NCCL (PR #1462)
  - OSU + NCCL for CUDA 12.8.0 (PR #1462) and for CUDA 12.6.0 (PR #1401)
  - Brought Zen5 on par with other targets w.r.t. CUDA packages (PR #1446)
  - SPH-EXA w/ CUDA 12.9.1 (PR #1453)

[Caspar, Bob]

Improved NVIDIA GPU support



Software-layer-scripts

- Fixed runtime CUDA version check (issue #189)
    - Runtime check effectively checked your *hardware*'s CUDA capability, but should have checked your driver
    - Typically passed because the '>' comparison was done on strings rather than numbers…
    - Fixed, and made CUDA driver version check dynamic, which means admins don't need to rerun `link_nvidia_host_libraries.sh` after every driver update (PR #202)
- Change 10.0f to 10.0 and 12.0f to 12.0 for CUDA 12.8.0 (doesn't support f-suffix yet, PR #200, #209)
- Fixed CUDA driver location check (PR #185)
- Skip installation of cuDNN >=9.12.0 when targeting CUDA Compute Capability 7.0, as it's not supported (PR #170)

[Caspar, Bob]

AMD GPU support



EasyBuild

- ROCm based toolchains defined PR #5099 (framework)
- ROCmComponent easyblock PR #3861
- 35 easyconfigs with ROCm components for ROCm 6.4.1 PR #25576
- Work in progress: defining higher level ROCm-based toolchains (rompi, rfoss, …)

[Caspar, Bob]

AMD GPU support

EESSI

- First successful tests with EESSI-extend, built ROCm-enabled GROMACS @ LUMI

Software-layer-scripts:

- Add AMD GPU support to EESSI-extend (PR #206, PR #218)
- Modify architect to allow overriding AMD GPU architectures (PR #215)
- Update EasyBuild hook that checks if GPU software is installed in GPU-prefixes (PR #217)

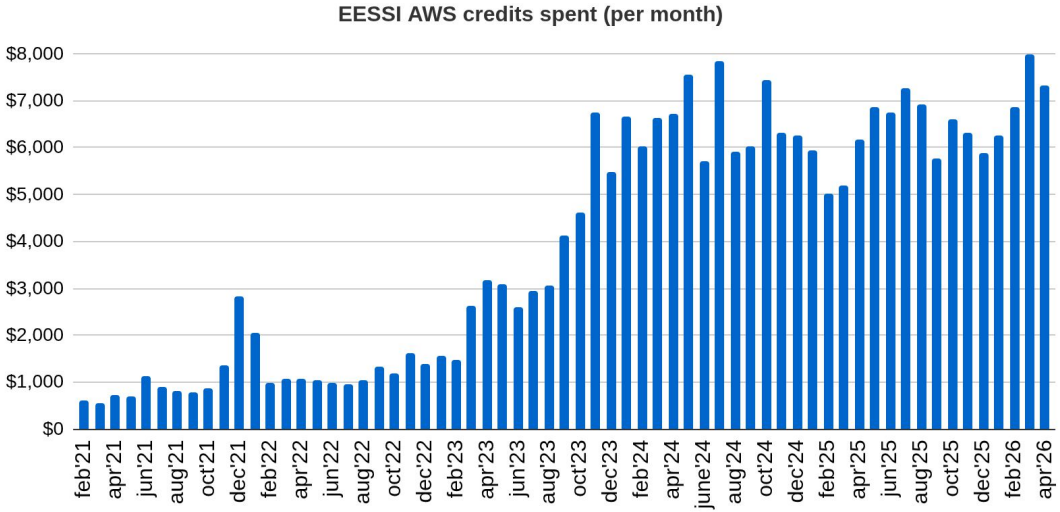
Software-layer:

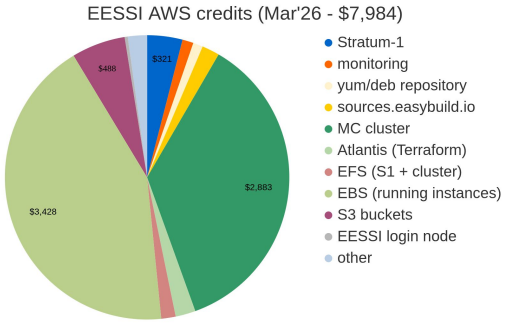
- ROCm-LLVM (PR #1473)
- Decided on optimization targets: gfx908, gfx90a, gfx942, gfx1030, gfx1100, gfx1101, gfx1200, gfx1201

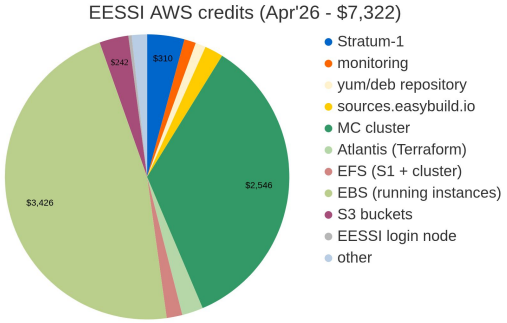
[Caspar, Bob]

Sponsored AWS credits

- ~$7,984 + $7,322 “spent” in Mar'26 + Apr'26 on Stratum-1 servers, monitoring, demos, sources.easybuild.io, Slurm clusters (build bot)
- **Bulk of consumed credits due to EESSI build-and-deploy bot**
- **Setting up new build cluster in Ireland (eu-west-1)**
- Regular sync meetings with AWS
- Received a new (big) batch of credits for 2026 🎉







[Kenneth, Bob, Alan]

EESSI Governance - (interim) Steering Committee

- **Governance for EESSI has been formalized in the EESSI documentation**
  - https://eessi.io/docs/governance
  - Feedback is still very welcome! A procedure is in place to consider (and vote on) changes
- Current members of EESSI Steering Committee (ESC):
  Kenneth (UGent) - chair, Bob (UGroningen), Caspar (SURF), Thomas (UBergen), Helena (Do IT Now)
- Most recent ESC meeting was held on 19 Feb 2026, next one planned for Tue 12 May 2026
- Agenda items of last ESC meeting:
  - Short-term roadmap for EESSI
  - Identifying and eliminating Single-Points-of-Failure (SPOFs)
  - Briefly: EuroHPC funding, EESSI “releases”, …

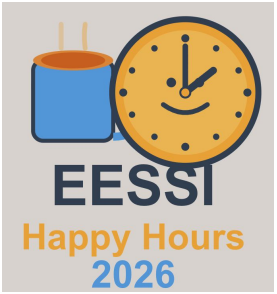
[Kenneth]

EESSI Happy Hour sessions ⏰ **Mondays 14:00-15:00** CE(S)T

EESSI Happy hour sessions 2026:

- Started Monday January 12 with EESSI Roadmap 2026

- 14 sessions so far in 2026
  - Topic series: EESSI Fundamentals Refresher
  - Topic series: What's New in EESSI 2025.06 ?
  - Topic series: Revisiting Ways to Add Software to EESSI
  - Topic series: CI/CD Workflows with EESSI



- Mondays between April–June 2026 will be dedicated to the EESSI webinar series

- EESSI Happy hour sessions will resume on June 8
  - Upcoming sessions will cover site specific topics

https://eessi.io/docs/training-events/happy-hours-sessions/

[Richard]

Webinar series on EuroHPC Federation Platform

- https://my-eurohpc.eu/training
- Via Zoom, open webinars, **registration required**
- Recordings available for 4 webinars:
  - Introduction to EFP,
  - **Federated Software Catalog (based on EESSI)**
  - EFP Interactive
  - EFP Authentication & Authorization Infrastructure (AAI)
- One more webinars to come:
  - Thu 11 June’26 14:00 CEST: EFP Workflows



[Kenneth, Alan]

11th EasyBuild User Meeting

easybuild.io/eum26

- Tue-Thu 21-23 April 2026 @ Guimarães, Portugal (home of Deucalion)
- **3-day event, day 3 focussed on EESSI**
- Blog posts up for all 3 days available at blog.easybuild.io
- YouTube playlist available for entire event: youtube.com/playlist?list=PLhnG…
  - Slides are available via this link





[Kenneth, Alan, …]

ISC'26 conference

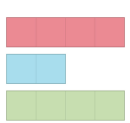




- https://isc-hpc.com
- 22-26 June 2026 @ Hamburg, Germany
- Proposal for half-day tutorial on EESSI accepted 🚀🚀🚀 (Mon 22 June 2026)
- Birds-of-a-Feather sessions on both EESSI & EuroHPC Federation Platform (EFP)
  - EFP: Wed 24 June at 10:45 | EESSI: Thu 25 June at 09:00
- Talks/demos on EESSI at various booths at ISC exhibit (to be confirmed)
- Most likely also some kind of social meetup… 🍻

[Kenneth]

EESSI/EasyBuild/Spack hackathon







- In-person training/hackathon on EasyBuild, EESSI, and Spack
- 19-21 May 2026 @ Bologna, Italy (CINECA)
- Tuesday is introductions to the tools and projects (EasyBuild, Spack, EESSI, EFP)
- Wednesday is short presentations on CI-related topics and hands-on
- Thursday is Choose-Your-Own-Adventure with expert support
- Remote attendance on Tue+Wed possible via Zoom: https://event.ugent.be/registration/ebeessispackhackathonmay26

[Kenneth, Alan, Lara, Davide]

EESSI webinar series (spring 2026)



- **Save the date!** Each webinar will start at 14:00 CEST
  - Mon 27 Apr'26: *Introduction to EESSI* - Helena ✅
  - Mon 4 May: *Building software on top + adding software to EESSI* - Kenneth + Lara ✅
  - Mon 11 May: *Using EESSI for Continuous Integration (CI)* - Alan
  - Mon 18 May: *Introduction to CernVM-FS* - Valentin Volkl
  - Monday 1 June: Using EESSI as a base for a central software stack



- More info (incl. registration) available via https://eessi.io/docs/training-events
- A separate webinar series is being planned for EasyBuild (June'26 ?)

EESSI mailing list eessi@list.rug.nl

- Low-frequency mailing list
  - Announcing bi-monthly meetings
  - Important updates and announcements
- To (un)subscribe, send an email to listserv@list.rug.nl with the following body:
  - Subscribe:

    ```
    SUBSCRIBE EESSI Firstname Lastname
    ```

  - Unsubscribe:

    ```
    UNSUBSCRIBE EESSI
    ```

Next EESSI update meetings

- EESSI update meetings are **bi-monthly** (odd months)
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 2 July 2026 14:00 CEST (12:00 UTC)
  - Thu 3 Sept 2026 14:00 CEST (12:00 UTC)
  - Thu 5 Nov 2026 14:00 CET (13:00 UTC)